Data Structures & Algorithms

Practical Sessions

LAB SESSION #5

Implement binary search algorithm

WHAT IS BINARY SEARCH?

- Binary Search is a searching algorithm for finding an element's position in a sorted array.
- In this approach, the element is always searched in the middle of a portion of an array.

Binary search can be implemented only on a sorted list of items. If the elements are not sorted already, we need to sort them first.

HOW BINARY SEARCH WORKS?

Binary Search Algorithm can be implemented in two ways which are discussed below.

- Iterative Method
- Recursive Method

The general steps for both methods are discussed below.

HOW BINARY SEARCH WORKS?

Initial Array



Suppose we need to find x=4

HOW BINARY SEARCH WORKS?

Set two pointers low and high at the lowest and the highest positions respectively.



HOW BINARY SEARCH WORKS?

Find the middle element mid of the array ie. arr[(low + high)/2]

=> arr[3] = 6



HOW BINARY SEARCH WORKS?

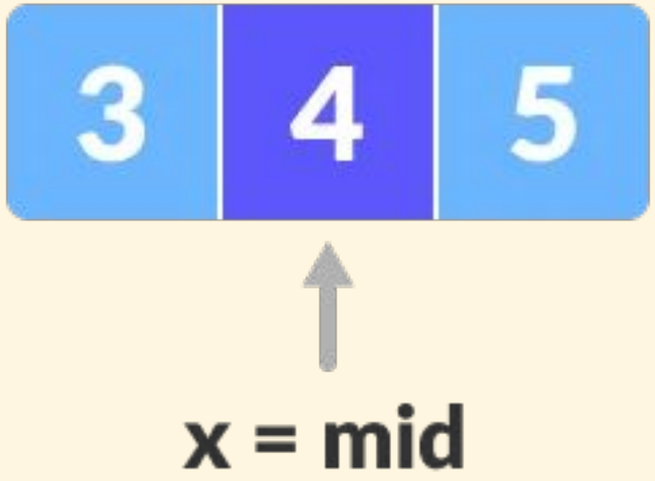
- If x == mid, then return mid. Else, compare the element to be searched with mid.
- If x > mid, compare x with the middle element of the elements on the right side of mid. This is done by setting low to low = mid + 1.
- Else, compare x with the middle element of the elements on the left side of mid. This is done by setting high to high = mid - 1.

HOW BINARY SEARCH WORKS?



HOW BINARY SEARCH WORKS?

- Repeat steps 3 to 6 until low meets high.
- or when x=4 is found



Algorithm: BINSEARCH(DATA, ITEM)

1. Set BEG = 0, END = N-1 and MID = INT((BEG + END) / 2)

2. Repeat steps 3 and 4 while BEG <= END AND DATA[MID] != ITEM

3. If ITEM < DATA[MID], then Set END = MID – 1, Else Set BEG = MID + 1

4. MID = INT((BEG + END) / 2)

5. If DATA[MID] = ITEM, then Set LOC = MID Else Set LOC = NULL

6. Exit